

# ENGINEERING SPECIFICATIONS

|                                                              |                      |                         |
|--------------------------------------------------------------|----------------------|-------------------------|
| <b>Document Name: IO list for KWK Siemens Double Circuit</b> |                      |                         |
| <b>Document No: ESP-23-24-009</b>                            | <b>Revision: R01</b> | <b>Date: 03/05/2023</b> |
| <b>Prepared By: SAM</b>                                      |                      | <b>Approved By: CGH</b> |

|                            |               |     |                                                               |
|----------------------------|---------------|-----|---------------------------------------------------------------|
| <b>Analog Input/Output</b> | <b>POL688</b> | X1  | Suction Pressure Comp 1                                       |
|                            |               | X2  | Discharge Pressure Comp-1 (4-20 mA)                           |
|                            |               | X3  | Oil Feed Pressure Comp 1(4-20 mA)                             |
|                            |               | X4  | Evaporator Water In Temperature                               |
|                            |               | X5  | Evaporator Water Out Temperature                              |
|                            |               | X6  | SPARE                                                         |
|                            |               | X7  | Suction Temperature                                           |
|                            |               | X8  | Discharge Temperature Comp 1                                  |
|                            |               | X9  | Liquid Line Temperature                                       |
|                            |               | X10 | Condenser Water In Temperature                                |
|                            |               | X11 | Condenser Water Out Temperature                               |
|                            | <b>POL985</b> | X1  | EXV Economiser 1(VFD command Comp 1 )(Analog Output 0-10 Vdc) |
|                            |               | X2  | Economiser EXV Comp1(Analog Output 0-10 Vdc)                  |
|                            |               | X3  | Desuperheater Out Temperature                                 |
|                            |               | X4  | EXV Command comp 1(Analog Output 0-10 Vdc)                    |
|                            |               | X5  | Oil Discharge Pressure Comp 1                                 |
|                            |               | X6  | Evaporator Flow Meter                                         |
|                            |               | X7  | Condenser Flow Meter                                          |
|                            |               | X8  | Economiser Out Pressure Comp 1                                |
|                            |               | B1  | Economiser Out Temperature Comp 1                             |
|                            |               | B2  | Sump Oil Temperature                                          |
|                            |               | B3  | Desuperheater In Temperature                                  |
|                            | <b>POL985</b> | X1  | Discharge Pressure Comp 2 ( 4-20 mA)                          |
|                            |               | X2  | Oil Feed Pressure Comp 2 ( 4-20 mA )                          |
|                            |               | X3  | Comp 2 Current (0-5 Vdc)                                      |
|                            |               | X4  | Oil Discharge Pressure Comp 2 ( 4-20 mA )                     |
|                            |               | X5  | VFD command Comp 2 (Analog Output 0-10 Vdc)                   |

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|  |    |                                                |
|--|----|------------------------------------------------|
|  | X6 | Economiser EXV Comp 2 (Analog Output 0-10 Vdc) |
|  | X7 | Economiser Out Pressure Comp 1 ( 4-20 mA )     |
|  | X8 | EXV Command comp 2 (Analog Output 0-10 Vdc)    |
|  | B1 | Discharge Temperature Comp 2                   |
|  | B2 | Economiser Out Temperature Comp 1              |
|  | B3 | SPARE                                          |

|                      |        |     |                                                 |
|----------------------|--------|-----|-------------------------------------------------|
| <b>Digital Input</b> | POL688 | D1  | Evaporator Flow switch                          |
|                      |        | D2  | Condenser Flow Switch                           |
|                      |        | DU1 | Comp 1 ON/OFF Switch (24V AC/DC)                |
|                      |        | DU2 | Voltage Protection Device/VFD Fault (24V AC/DC) |
|                      |        | DL1 | Transition Feedback Comp 1 (AC 115...230 V)     |
|                      |        | DL2 | Transition Feedback Comp 2(AC 115...230 V)      |
|                      | POL985 | D1  | Oil Level Switch                                |
|                      |        | D2  | External Emergency Stop                         |
|                      |        | D3  | Main On Off Switch                              |
|                      |        | DL1 | Winding Protection Comp 1 (AC 115...230 V)      |
|                      |        | DL2 | Winding Protection Comp 2 (AC 115...230 V)      |
|                      | POL985 | D1  | Comp 2 ON/OFF Switch                            |
|                      |        | D2  | Voltage Protection Device/VFD Fault Comp 2      |
|                      |        | D3  | ANTIFREEZE SWITCH                               |
|                      |        | DL1 | LP Cutout (AC 115...230 V)                      |
|                      |        | DL2 | HP Cutout (AC 115...230 V)                      |

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|                       |        |     |                                          |
|-----------------------|--------|-----|------------------------------------------|
| <b>Digital Output</b> | POL688 | Q1  | Compressor (Delta) Comp/VFD Latch Comp 2 |
|                       |        | Q2  | Compressor (Delta) Comp/VFD Latch Comp 1 |
|                       |        | Q3  | Compressor (Main) Comp 1                 |
|                       |        | Q4  | Compressor (Main) Comp 2                 |
|                       |        | Q5  | Oil Feed Solenoid Comp 1                 |
|                       |        | Q6  | Oil Feed Solenoid Comp 2                 |
|                       |        | Q7  | SPARE                                    |
|                       |        | Q8  | SPARE                                    |
|                       |        | DO1 | SPARE                                    |
|                       |        | DO2 | SPARE                                    |
|                       | POL985 | Q1  | SPARE                                    |
|                       |        | Q2  | Cooling Tower Potential Free             |
|                       |        | Q3  | Condenser Pump Potential Free            |
|                       |        | Q4  | Evaporator Pump Potential Free           |
|                       |        | Q5  | Comp 1 Alarm Potential Free              |
|                       |        | Q6  | Comp 2 Alarm Potential Free              |
|                       |        | Q7  | Oil Heater Relay                         |
|                       |        | Q8  | Oil Pump Comp 1                          |
|                       |        | DO1 | SPARE                                    |
|                       |        | DO2 | SPARE                                    |
|                       | POL985 | Q1  | FWD-ATCS                                 |
|                       |        | Q2  | REV-ATCS                                 |
|                       |        | Q3  | SPARE                                    |
|                       |        | Q4  | SPARE                                    |
|                       |        | Q5  | SPARE                                    |
|                       |        | Q6  | SPARE                                    |

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|  |     |                                           |
|--|-----|-------------------------------------------|
|  | Q7  | SPARE                                     |
|  | Q8  | Oil Pump Comp 2                           |
|  | DO1 | Loading Coil (V1) Potential Free Comp 2   |
|  | DO2 | Unloading Coil (V2) Potential Free Comp 2 |

|                          |                               |
|--------------------------|-------------------------------|
| Controller T-IP Ethernet | PVR 7 INCH Communication Port |
| POL688                   | Main Board                    |
| POL985                   | Expansion Board               |

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